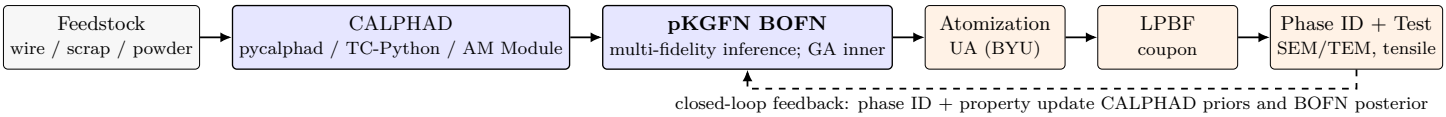
**AMMTO Concept Outline — FLAIME-CM: Domestic-Feedstock Printable Aluminum Alloys, with the Critical Materials Innovation Hub**

**PI:** Sterling Baird (Brigham Young University, Materials Science & Engineering). **Solicitation:** DOE EERE AMMTO Multi-Topic FOA / CMI Hub partnership track. **Type:** 2–3 yr, ∼$1–3M (AMMTO Multi-Topic envelope). **Keywords:** critical materials substitution, LPBF, ultrasonic atomization, BOFN, CALPHAD, TRL-4.




**Problem**

Printable Al alloy families on the AMMTO trajectory rely on Co/Ni/Sc/REE additions; substitution toward domestically abundant Mn/Cr/Zr/Si/Fe streams requires a discovery loop that explicitly carries a feedstock supply-risk objective alongside printability and elevated-temperature strength. **Mapping to the 2023 DOE Critical Materials List:** the substitution targets directly displace Co (Critical 2025–2035), Sc/REE (Critical), and Ni (Near-Critical) by design — each candidate alloy is evaluated against a per-element exposure score traceable to the current DOE list, scored by the CMI Hub. The Al–Ce trajectory (CMI Hub → Eck Industries → Magna) shows a CMI-coupled discovery→validation pipeline can reach commercial deployment; no analogous trajectory yet exists for the Al–Mn–Cr–Zr / Al–Zr / Cu–Cr–Zr families that the FLAIME loop is positioned to address.

**Idea, deliverable cadence, and go/no-go**

A 2–3 yr project applying the FLAIME discovery loop — pKGFN BOFN over CALPHAD multi-fidelity nodes (`pycalphad`, TC-Python/TCAL10, AM Module), validated by ultrasonic atomization (UA) and LPBF — to one alloy family per project year. Baseline: Al–Mn–Cr–Zr and Al–Zr. Stretch (Y2–Y3): Cu–Cr–Zr or Be-substitution. **Hypothesis:** a supply-risk-aware BOFN–CALPHAD loop, with CMI-supplied feedstock data and Ames cross-check production runs, can deliver one TRL-4 candidate alloy and an associated UA feedstock specification per project year. **Annual go/no-go:** (i) one validated candidate alloy with UA powder spec, LPBF parameter window, and tensile data at room and elevated temperature; (ii) CMI-scored supply-risk reduction relative to the matched Al–Ce/Al–Ni–Sc benchmark; (iii) Ames cross-check print confirming reproducibility outside BYU.

**Cost-share and BABA compliance**

The 20% statutory non-federal cost-share will be met by documented in-kind contributions from Valimet (gas-atomized comparator powder, characterization time) and AMAZEMET (UA hardware engineering support), with BYU committing residual cost-share through PI summer salary and HPC; valuation, allowability, and PoP documentation will be audit-ready. **BABA:** the manufactured articles funded under this assistance are the UA-derived powder feedstock and the LPBF-produced Al alloy coupon/tested article; both are produced in the U.S. on the BYU-installed AMAZEMET rePOWDER and on U.S.-based LPBF systems. AMAZEMET's role in cost-share is upstream engineering support (process know-how transfer), not procurement of foreign-manufactured end product; no foreign-manufactured iron, steel, or manufactured product is incorporated into the deliverable article stream. If contracting determines a waiver is needed, Valimet provides a fully domestic atomization comparator path. Any EU mirror-site engineering interaction is knowledge exchange only and will be disclosed under research-security review.

**Partnership structure and downstream-adopter pathway**

**Funded:** BYU (Baird, lead — discovery loop, in-house UA powder synthesis, LPBF coordination); CMI Hub (supply-chain modeling, supply-risk scoring, reference validation prints under separate CMI funding). **Collaborator with own funding:** Ames Laboratory (cross-check production runs; per AMMTO/EERE convention Ames brings its own DOE funding rather than a budget line through BYU). **Industrial:** Valimet, AMAZEMET (cost-share in-kind), Thermo-Calc (CALPHAD). **Downstream adopter (gap acknowledged, in-progress):** no Eck/Magna-class transition partner is yet under letter for Al-MPEA; before the full white paper we will close one of two paths — LPBF service bureau adopting the qualified powder/parameter window, or automotive/aerospace tier-1 evaluator analogous to Magna's Al-Ce role — and disclose the named partner.

**Preliminary results (FLAIME pipeline, BYU, Apr 2026)**

UA composition modification from AlSi10Mg + Al wire + Si feed confirmed by SEM/EDS; Thermo-Calc AM Module simulations on candidate FLAIME compositions; CALPHAD jobs on BYU HPC. AlSi10Mg LPBF tensile baseline; elevated-temperature tensile data on a standard Al specimen; working pKGFN BOFN; automated powder dispensing demonstrated. The Ames letter of support is on file. *Y1 risk-retiring milestone:* one Al-Mn-Cr-Zr or Al-Zr LPBF coupon set produced via the proposed feedstock route, with printability evidence, room-temperature tensile vs. AlSi10Mg, and a per-element exposure-score delta vs. an Sc/REE-bearing comparator — the closed-loop result reviewers will want before TRL-4 claims.

**Scope separation from pending DOE Genesis Mission Topic 3B and request**

**Task-level scope separation.** *Genesis tasks (DOE / Topic 3B):* alloy-agnostic platform construction — hardware integration, scheduler, data fabric, autonomous orchestration, general platform validation. *AMMTO tasks (this proposal):* (T1) per-alloy-family campaign design with explicit critical-materials-list mapping; (T2) UA powder production and qualification; (T3) LPBF parameter-window development; (T4) elevated-temperature mechanical testing to TRL-4; (T5) CMI-scored supply-risk benchmarking and Ames cross-check. Personnel labor is itemized per task; data packages and reporting artifacts are alloy-family-specific and not Genesis-platform deliverables. No budget line, equipment purchase, or scheduler/data-fabric task appears in both. **Request:** a scoping call with the cognizant AMMTO PM and the CMI Hub director (via the Ames letter contact) to confirm the per-element exposure score and TRL-4 framing match the current Critical Materials List before we draft a full white paper.

*Repo:* https://github.com/vertical-cloud-lab.